Nagar YuwakShikshanSanstha's

YeshwantraoChavan College of Engineering

(An Autonomous Institution affiliated to RashtrasantTukadojiMaharaj Nagpur University)

(Accredited 'A' Grade by NAAC)

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Form No YI-04

Department of Electronics Engineering

STUDENT'S DIARY/ LOG

Your Name: Khushi Narendra Atara

Your Institute: Yeshwantrao Chavan College of Engineering Nagpur

Company where internships are carried out : Cognizant Technology Solutions Pvt. Ltd

Period: From: 31/01/25 To: 13/05/26

Week: 11		DATE: From	13/04/26 to 17/04/26
Dept./Division	Programmer Analyst Trainee	Name of Project	Training and Internship Program
Name of HOD/ Supervisor With e-mail id		Name: Ashly M. Benny	
		Email Id: ashly.bennym@cognizant.com	
Main points of the week		Stage 2 – Automation Framework & DevOps: GIT, Maven, Selenium Grid, Jenkins (Days 46–51)	

• Studied Automation Framework concepts: Data-Driven, Keyword-Driven, Page Factory, Hybrid Framework design combining POM + Data-Driven + Keyword-Driven approaches.

• Learned Test Automation Framework components: Test Data Handling (Excel, CSV, XML, JSON), Test Report and Test Log generation using Extent Reports and Allure Reports.

• Understood Custom Logging with Log4j in Selenium and Screenshot Capture Strategies for failure analysis.

• Learned GIT and GitHub: basic commands (clone, add, commit, push, pull), branching and merging strategies, and pushing automation framework to remote repository.

• Studied Maven: Introduction, creating and managing Maven projects, Maven Build Lifecycle (clean, compile, test, package), managing dependencies and plugins, and executing tests using Maven goals.

• Understood Jenkins CI: setting up Jenkins CI server, creating Jenkins Pipeline, integrating GIT and Maven projects, and scheduling/triggering automation runs.

• Studied Selenium Grid: architecture (Hub and Node), setting up and configuring Grid, and running tests in parallel on multiple browsers.

• Completed mandatory Tekstac hands-on: Git Config, Clone Repo, Add/Commit/Push, Pull and Merge, Merge Resolve Conflict, Git Tags, Build Maven Project, Pipeline compile and test, Build Web Application.

• Completed additional hands-on: GIT installation and commands implementation, Install Jenkins and Creation of new job.

Signature and stamp of Industry Supervisor

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Fortnightly Progress Report

(To be mailed every alternate Monday to Project Guide, designated person(s) of the company, your HoD and the concerned Dean from your institute)

1. Company Overview:

Cognizant Technology Solutions is a leading global IT services and consulting company headquartered in the USA, with a major delivery presence across India. It specializes in digital transformation, cloud computing, AI, automation, software engineering, and quality assurance. Cognizant's QEA (Quality Engineering and Assurance) division focuses on intelligent and automated quality assurance, modernisation assurance, and experience assurance for enterprise clients across healthcare, finance, and technology sectors.

2. Who did you interact with in this period? Which departments did you visit/work in?

During Week 11, I interacted with Cognizant Academy trainers and fellow GenC trainees during Master Sessions on Automation Framework design and DevOps tools. All sessions were conducted within the Training division. The week covered Day 46 (Automation Framework) and Days 47 to 51 (DevOps – GIT, Maven, Grid, Jenkins), with hands-on exercises being the primary activity alongside daily trainer-led sessions.

3. List down (maximum) three activities carried out by you in this period. Describe each one briefly.

Activity 1 – Automation Framework Design (Day 46): Attended the Master Session on designing a complete Hybrid Automation Framework combining POM, Data-Driven, and Keyword-Driven approaches. Studied Test Data Handling using Excel/CSV/XML/JSON, Extent Reports, Log4j logging, and screenshot capture strategies for failure analysis.

Activity 2 – GIT & Maven Hands-On (Days 47–51): Practiced core GIT commands (clone, add, commit, push, pull, merge, resolve conflicts, tags) on the Tekstac platform. Also learned Maven project structure, dependency management via pom.xml, Maven build lifecycle, and executing Selenium tests using Maven goals from the command line.

Activity 3 – Jenkins CI & Selenium Grid Setup: Configured Jenkins CI server, created a Jenkins Pipeline integrating GIT and Maven, and set up job scheduling for automated nightly test runs. Also studied Selenium Grid Hub and Node architecture and practiced parallel test execution across multiple browsers.

4. What problems did you face and what did you do to address them? What was the learning that resulted from each of them?

Faced difficulty resolving merge conflicts in GIT when multiple branches had changes to the same file. Practiced the Merge – Resolve Conflict hands-on on the Tekstac platform and referred to BU Chapter 17 content for guidance. This experience taught the importance of frequent commits and clear branching strategies to minimize merge conflicts in team-based development environments.

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5. What process(s)/system(s) did you familiarize yourself with/study as part of your work or out of your interest?

Familiarized myself with GIT version control system and GitHub for remote repository management, Maven for Java project build automation and dependency management, Jenkins for CI/CD pipeline creation and test scheduling, and Selenium Grid for distributed parallel test execution. Also studied BU Training Content Chapters 17 to 19 covering Selenium Grid, Maven, GIT Version Control, and Jenkins CI Systems with their PPT and video reference content.

6. If you have seen anything where an improvement is possible/required, please mention/describe that here.

A dedicated hands-on session showing the complete end-to-end CI/CD pipeline – from a developer pushing code to GIT, Maven building the project, and Jenkins automatically triggering the Selenium test suite and generating reports – would greatly help trainees understand how all these DevOps tools work together in a real enterprise pipeline rather than learning each tool in isolation.

7. What did you find novel/exciting/valuable during this period?

Setting up a Jenkins pipeline that automatically triggers Selenium tests whenever code is pushed to GitHub was the most exciting experience of this week. Seeing the entire CI/CD workflow – from code commit to test execution to report generation – happen automatically without any manual intervention gave a real sense of how modern DevOps-driven quality engineering works in enterprise projects at Cognizant.

8. What according to you, can be of use to your institute and students and in what ways?

Introducing GIT, Maven, and Jenkins as part of the academic software engineering or final-year project curriculum would be immensely valuable. Most students submit projects without any version control, build automation, or CI practices. Even a basic lab session covering GIT branching, Maven project setup, and Jenkins job creation would make students significantly more industry-ready compared to their peers.

9. Have you been able to complete everything that you had planned last fortnight? If not, what have you not completed and why?

Yes, all planned activities for Week 11 were completed. The Automation Framework session (Day 46), mandatory and additional Tekstac hands-on exercises for GIT and Maven (Days 47–51), BU Chapter content (Chapters 17–19), Udemy Section 8 (CI – Maven, Git, GitHub & Jenkins), and the Integrated Project deliverable for Scalable Execution Infrastructure were all progressed within the scheduled time.

Khushi N. Atara

Name and Signature of Student.

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